DWEEP VARTAK

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Experience

Quidich Innovations Labs - Machine Learning Intern ***Jan'2025- Mar'2025***

Worked on AI solutions for live sports broadcasting, with technologies featured in the ICC T20 World Cup Finals and IPL 2025, enhancing viewer experience through real-time overlays and player analytics.

- Developed deep learning models for **player, head, and body detection** powering **Stiqy** overlays and **Quidich Tracker**, achieving **92% detection accuracy** across diverse broadcast scenarios.
- Managed data annotation efforts and automated preprocessing workflows, reducing manual workload and enabling overnight training cycles.
- Monitored model performance using **Weights & Biases** and **Voxel51**, enabling detailed result tracking, metric visualization, and iterative tuning for deployment readiness.

Technologies: Python, PyTorch, YOLO, OpenCV, Label Studio, WandB, Voxel51, Linux

Projects

Edge AI–Enabled Wearable Assistive Device for the Visually Impaired ***Dec'25***

Developed a real-time wearable Edge AI system integrating computer vision, speech processing, and embedded inference for low-latency, on-device perception and navigation assistance, with a strong focus on hardware–software co-design and system optimization.

- Designed and built a **real-time wearable vision system** using **Raspberry Pi 5 + Hailo-8L AI accelerator** for low-latency edge inference.
- Built a multimodal Edge AI pipeline integrating computer vision, speech processing, and NLP for **real-time navigation, scene understanding, and visual question answering**.
- Achieved **sub-second response times** using **model quantization, ONNX optimization, and multi-threaded pipelines**.
- Technologies Used: Raspberry Pi 5, Hailo-8L, Python, PyTorch, ONNX Runtime, OpenCV, YOLO, Depth Estimation Models, Speech-to-Text, Text-to-Speech, Multithreading, Embedded Linux

Flappy Bird Reinforcement Learning Agent ***Sept'24***

Developed a reinforcement learning agent to play Flappy Bird using computer vision and deep learning techniques.

- Created a custom environment using Gymnasium and MSS for real-time screen capture and state observation.
- Conducted extensive testing, resulting in the agent ***scoring up to 176 points in a single session***.
- Technologies Used: Python, OpenCV, Gymnasium, Stable Baselines3, PaddleOCR, Deep Q-Networks (DQN).

Intelligent Traffic Flow Optimization using OpenCV ***Feb'23***

Led the development of an intelligent traffic management solution leveraging computer vision with OpenCV. Implemented real-time vehicle detection, tracking, and congestion analysis to optimize traffic flow.

- Demonstrated improved traffic efficiency with a ***23.10% increase*** in the simulated passage of cars through the intersection.
- Technologies Used: Python, OpenCV, Computer Vision, Image Processing

Skills

- **Programming & Systems:** C, C++, Python, Java, Linux, Bash, Multithreading, Multiprocessing
- **Computer Vision & Deep Learning:** OpenCV, YOLO, PyTorch, TensorFlow, Object Detection, Tracking, OCR, Depth Estimation, Reinforcement Learning (DQN), Vision-Language Models (VLM)
- **Edge AI & Embedded Systems:** Raspberry Pi, Edge AI Deployment, ONNX Runtime, Model Quantization, Embedded Linux, Real-Time Inference Pipelines
- **Data & ML Tools:** Scikit-learn, Pandas, NumPy, Matplotlib, Weights & Biases, Voxel51, Label Studio
- **Backend & DevTools:** Flask, SQL, Git, GitHub
- **Soft Skills:** Problem Solving, Team Collaboration, Adaptability, Time Management

Education

K. J. Somaiya Institute of Technology, Mumbai **2021 - 2025**

- B.Tech in Artificial Intelligence and Data Science | CGPA: ***9.23***

Gaurav High School, Navi Mumbai **2019 - 2021**

- *HSC (Class XII), Percentage: 90.0%*

I.E.S. J.N.P.Vidyalaya, Navi Mumbai **2018 - 2019**

- *SSC (Class X), Percentage: 90.6%*